ENGINEERING MATHEMATICS

UNIT–II : TRIGONOMETRY

(As per DCET C20 Diploma Curriculum — Marks: 4)

DCET Exam Note:

Syllabus Coverage:

- Concept of angles, their measurement, Radian measure and related conversions
- Signs of trigonometric ratios in different quadrants (ASTC rule)
- Trigonometric ratios of allied angles ($90^\circ \pm \theta$, $180^\circ \pm \theta$, $270^\circ \pm \theta$, $360^\circ \pm \theta$)
- Trigonometric ratios of compound angles (without proof)
- Trigonometric ratios of multiple angles
- Transformation formulae

1. CONCEPT OF ANGLES AND MEASUREMENT

Angle Definition

An angle is formed by rotation of a ray from an initial position to a terminal position about a fixed point called vertex.

1.1 Measurement Systems

Formula:

Degree System: 1 complete revolution = 360°

Radian System: 1 complete revolution = 2π radians

1.2 Conversion Formulas

Formula:

Degrees to Radians: $\theta_{\text{rad}} = \theta_{\text{deg}} \times \frac{\pi}{180}$

Radians to Degrees: $\theta_{\text{deg}} = \theta_{\text{rad}} \times \frac{180}{\pi}$

Illustration 1.1

Convert 60° to radians.

Solution:

$$60^\circ = 60 \times \frac{\pi}{180} = \frac{\pi}{3} \text{ radians}$$

Illustration 1.2

Convert $\frac{3\pi}{4}$ radians to degrees.

Solution:

$$\frac{3\pi}{4} = \frac{3\pi}{4} \times \frac{180}{\pi} = 135^\circ$$

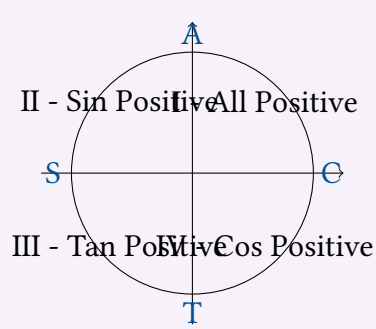
2. STANDARD TRIGONOMETRIC RATIOS

Function	Symbol	Definition
Sine	$\sin\theta$	$\frac{\text{Opposite}}{\text{Hypotenuse}}$
Cosine	$\cos\theta$	$\frac{\text{Adjacent}}{\text{Hypotenuse}}$
Tangent	$\tan\theta$	$\frac{\text{Opposite}}{\text{Adjacent}}$
Cosecant	$\csc\theta$	$\frac{1}{\sin\theta}$
Secant	$\sec\theta$	$\frac{1}{\cos\theta}$
Cotangent	$\cot\theta$	$\frac{1}{\tan\theta}$

Table 1: Basic Trigonometric Ratios

3. SIGNS OF TRIGONOMETRIC RATIOS (ASTC RULE)

ASTC Rule - Remember as "All Students Take Calculus"



Quadrant	Range	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
I	0° to 90°	+	+	+	+	+	+
II	90° to 180°	+	-	-	+	-	-
III	180° to 270°	-	-	+	-	-	+
IV	270° to 360°	-	+	-	-	+	-

Table 2: Signs in Different Quadrants

Important:

Mnemonic: "All Silver Tea Cups" or "Add Sugar To Coffee"

All (I), **S**ine (II), **T**angent (III), **C**osine (IV)

4. TRIGONOMETRIC RATIOS OF ALLIED ANGLES

Allied Angles Concept

Angles that differ by multiples of 90° ($\frac{\pi}{2}$) are called allied angles.

4.1 Angles of form $90^\circ \pm \theta$ ($\frac{\pi}{2} \pm \theta$)

Formula:

$\sin(90^\circ - \theta) = \cos\theta \quad \sin(90^\circ + \theta) = \cos\theta$

$\cos(90^\circ - \theta) = \sin\theta \quad \cos(90^\circ + \theta) = -\sin\theta$

$\tan(90^\circ - \theta) = \cot\theta \quad \tan(90^\circ + \theta) = -\cot\theta$

$\csc(90^\circ - \theta) = \sec\theta \quad \csc(90^\circ + \theta) = \sec\theta$

$\sec(90^\circ - \theta) = \csc\theta \quad \sec(90^\circ + \theta) = -\csc\theta$

$\cot(90^\circ - \theta) = \tan\theta \quad \cot(90^\circ + \theta) = -\tan\theta$

4.2 Angles of form $180^\circ \pm \theta$ ($\pi \pm \theta$)

Formula:

$\sin(180^\circ - \theta) = \sin\theta \quad \sin(180^\circ + \theta) = -\sin\theta$

$\cos(180^\circ - \theta) = -\cos\theta \quad \cos(180^\circ + \theta) = -\cos\theta$

$\tan(180^\circ - \theta) = -\tan\theta \quad \tan(180^\circ + \theta) = \tan\theta$

4.3 Angles of form $270^\circ \pm \theta$ ($\frac{3\pi}{2} \pm \theta$)

Formula:

$\sin(270^\circ - \theta) = -\cos\theta \quad \sin(270^\circ + \theta) = -\cos\theta$

$\cos(270^\circ - \theta) = -\sin\theta \quad \cos(270^\circ + \theta) = \sin\theta$

$\tan(270^\circ - \theta) = \cot\theta \quad \tan(270^\circ + \theta) = -\cot\theta$

4.4 Angles of form $360^\circ \pm \theta$ ($2\pi \pm \theta$)

Formula:

$\sin(360^\circ - \theta) = -\sin\theta \quad \sin(360^\circ + \theta) = \sin\theta$

$\cos(360^\circ - \theta) = \cos\theta \quad \cos(360^\circ + \theta) = \cos\theta$

$\tan(360^\circ - \theta) = -\tan\theta \quad \tan(360^\circ + \theta) = \tan\theta$

Important:

Trick: Function changes to co-function for odd multiples of 90°

For $90^\circ \pm \theta$ and $270^\circ \pm \theta$: sin $\leftrightarrow$ cos, tan $\leftrightarrow$ cot, sec $\leftrightarrow$ csc

5. STANDARD TRIGONOMETRIC VALUES

θ	Radians	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
0°	0	0	1	0	∞	1	∞
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$	2	$\frac{2}{\sqrt{3}}$	$\sqrt{3}$
45°	$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1	$\sqrt{2}$	$\sqrt{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{2}{\sqrt{3}}$	2	$\frac{1}{\sqrt{3}}$
90°	$\frac{\pi}{2}$	1	0	∞	1	∞	0

Table 3: Standard Values (Must Memorize for DCET)

DCET Exam Note:

Memory Trick for sin values:

$\sin 0^\circ = \frac{\sqrt{0}}{2} = 0$, $\sin 30^\circ = \frac{\sqrt{1}}{2} = \frac{1}{2}$, $\sin 45^\circ = \frac{\sqrt{2}}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\sin 90^\circ = \frac{\sqrt{4}}{2} = 1$

6. COMPOUND ANGLE FORMULAS

Compound Angles

Angles of the form $A \pm B$, $A + B$, $A - B$ are compound angles.

6.1 Addition Formulas

Formula:

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$
$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$
$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

6.2 Subtraction Formulas

Formula:

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$
$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$
$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Illustration 6.1

Find $\sin 75^\circ$ using compound angle formula.

Solution:

$$\sin 75^\circ = \sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$$
$$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{3}+1}{2\sqrt{2}}$$

7. MULTIPLE ANGLE FORMULAS

7.1 Double Angle Formulas

Formula:

$$\sin 2\theta = 2\sin\theta\cos\theta$$
$$\cos 2\theta = \cos^2\theta - \sin^2\theta$$
$$\cos 2\theta = 2\cos^2\theta - 1$$
$$\cos 2\theta = 1 - 2\sin^2\theta$$
$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

7.2 Triple Angle Formulas

Formula:

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$
$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$$
$$\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1-3\tan^2\theta}$$

Illustration 7.1

If $\sin\theta = \frac{3}{5}$, find $\sin 2\theta$.

Solution:

$$\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{1-\frac{9}{25}} = \frac{4}{5}$$
$$\sin 2\theta = 2\sin\theta\cos\theta = 2\times\frac{3}{5}\times\frac{4}{5} = \frac{24}{25}$$

8. TRANSFORMATION FORMULAS

Transformation Formulas

These convert sum/difference to product and vice versa.

8.1 Product to Sum Formulas

Formula:

$\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$
$\cos A \sin B = \frac{1}{2}[\sin(A+B) - \sin(A-B)]$
$\cos A \cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$
$\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$

8.2 Sum to Product Formulas

Formula:

$\sin C + \sin D = 2\sin\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$
$\sin C - \sin D = 2\cos\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$
$\cos C + \cos D = 2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$
$\cos C - \cos D = -2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

Illustration 8.1

Express $\sin 75^\circ + \sin 15^\circ$ as product.

Solution:

$$\sin 75^\circ + \sin 15^\circ = 2\sin\left(\frac{75^\circ + 15^\circ}{2}\right)\cos\left(\frac{75^\circ - 15^\circ}{2}\right)$$

$$= 2\sin 45^\circ \cos 30^\circ = 2 \times \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{6}}{2}$$

Practice Problems (DCET Pattern)

DCET Exam Note:

Solve these typical DCET questions:

1. Convert 150° to radians. [Ans: $\frac{5\pi}{6}$]
2. Find $\sin 225^\circ$ using ASTC rule. [Ans: $-\frac{1}{\sqrt{2}}$]
3. Simplify: $\sin(180^\circ - \theta)\cos(90^\circ + \theta)$ [Ans: $-\sin^2\theta$]

4. If $\cos\theta = \frac{5}{13}$, find $\sin 2\theta$. [Ans: $\frac{120}{169}$]
5. Prove: $\frac{\sin 2\theta}{1+\cos 2\theta} = \tan\theta$
6. Express $\cos 4\theta$ in terms of $\cos\theta$. [Ans: $8\cos^4\theta - 8\cos^2\theta + 1$]
7. Simplify: $\sin 20^\circ \cos 10^\circ + \cos 20^\circ \sin 10^\circ$ [Ans: $\frac{1}{2}$]
8. Find $\tan 15^\circ$ using $\tan(45^\circ - 30^\circ)$. [Ans: $2 - \sqrt{3}$]

Quick Revision Points

- 1 radian = $\frac{180}{\pi}$ degrees
- ASTC: All, Sin, Tan, Cos
- For $90^\circ \pm \theta$: function $\rightarrow$ co-function
- $\sin^2\theta + \cos^2\theta = 1$
- $\sin 2\theta = 2\sin\theta\cos\theta$
- $\cos 2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$
- $\sin(A+B) = \sin A\cos B + \cos A\sin B$
- $\cos(A-B) = \cos A\cos B + \sin A\sin B$
- $\sin C + \sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$
- $\cos C - \cos D = -2\sin\frac{C+D}{2}\sin\frac{C-D}{2}$

DCET Exam Note:

DCET Exam Tips:

- Memorize standard values: $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$
- Practice ASTC rule for negative angles
- Use compound angles to find values like $75^\circ, 15^\circ, 105^\circ$
- Remember: $\sin^2\theta + \cos^2\theta = 1$ always
- For multiple angles: $\sin 2\theta$, $\cos 2\theta$ frequently asked
- Transformation formulas help simplify expressions
- Time management: 3-4 minutes per question

Trigonometry Formula Sheet

Basic Identities:

$$\sin^2\theta + \cos^2\theta = 1$$
$$1 + \tan^2\theta = \sec^2\theta$$
$$1 + \cot^2\theta = \csc^2\theta$$

Double Angle:

$$\sin 2\theta = 2\sin\theta\cos\theta$$
$$\cos 2\theta = 1 - 2\sin^2\theta$$
$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

Sum to Product:

$$\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$$
$$\cos A + \cos B = 2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$$

Product to Sum:

$$\sin A\cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$$
$$\cos A\cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$$